ANA VICTÓRIA LADEIRA

Senior AI Engineer & LLM Expert | RAG · Knowledge Graphs · MLOps · 12+ Years

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Senior AI Engineer with 12+ years delivering production AI systems across scientific research, government, and enterprise environments. Deep specialist in LLM pipelines, RAG architectures, and Knowledge Graphs, with a rare combination of hands-on engineering, research credibility (ESA, peer-reviewed publications), and product ownership. Experienced leading data teams and driving AI adoption at the organizational level. Available for senior AI consulting and contracting engagements.

EXPERIENCE

Data Lead

Portal de Compras Publicas

Sep 2025 - Current Brasilia, Brazil

- Architected end-to-end LLM document extraction pipelines for government procurement funding calls, reducing manual processing overhead for high-stakes public contracts.
- Lead Data team and organized deliverables for dashboards, machine learning models and other data products.
- Built and deployed classification algorithms with over 100k+ target classes.
- Driving AI strategy review across the organization, identifying optimization opportunities and aligning initiatives to measurable outcomes.
- Established a generative AI knowledge program for the department, accelerating adoption of LLMs and agent-based systems across teams.

Senior Data Scientist II

Elsevier

Jun 2024 - Sep 2025 Amsterdam, the Netherlands

- Modernized classification pipelines for 100M+ scientific publications, migrating from legacy systems to state-of-the-art LLM-based solutions with measurable quality improvements.
- Redesigned full MLOps lifecycle: automated monitoring, quality gates, retraining triggers, and production redeployment — reducing operational friction across the model fleet.
- Established cross-functional best practices between engineering and data science, including algorithmic governance frameworks and structured user feedback loops.
- Expanded ML pipelines to support multilingual scientific content, opening new customer segments and extending product reach beyond English-language markets.
- Lead departmental knowledge sharing initiatives on generative AI, LLMs, agents, and other recent developments in Data Science

Knowledge Management Engineer & Data Scientist

European Space Agency

Oct 2021 – Apr 2024 Noordwijk, the Netherlands

- Designed and deployed RAG pipelines and vector database architectures for internal document retrieval across ESA's knowledge base — among the earliest production RAG implementations in a scientific institution.
- Built Knowledge Graphs connecting heterogeneous data sources across the Agency, enabling expertise mapping and cross-project knowledge discovery at institutional scale..
- Contributed to working group on AI practices throughout the agency and how to harmonize different technology stacks across different projects.

EDUCATION

Masters in Data Science for Decision Making

Maastricht University

Feb 2019 – Oct 2021

- Master thesis topic: *A Bayesian Recommender System for Content Supporting the Formation of Healthy Habits*, written as part of partnership with Habtic.
- Projects: *Measuring Similarities Between Different Ancient Games to Support Ludoarcheologists* and Fitting a Family of Curves with Global and Local Parameters, to do curve fitting on energy curves from tests with transistors.

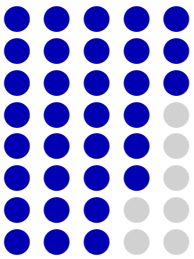
Bachelors in Data Science and Artificial Intelligence

Maastricht University

Sep 2016 – Jan 2019

- Part of honours program Knowledge Engineering @ Work, a 2 years internship with a local company (Accenture)

STRENGTHS



FRAMEWORKS

Python PydanticAI HuggingFace MLFlow Claude Code Openclaw n8n Anti Gravity Langchain Langgraph LlamaIndex Scikit-Learn Pyspark PyTorch Numpy NLTK PowerBI Pandas Gephi VSCode TensorFlow Vaticle TypeDB Neo4J Evidently

- Developed LLM-to-Knowledge Graph pipelines to automatically structure unstructured text (academic papers, procurement reports) into queryable graph databases.
- Organized LING4S - The 1st workshop on Computational Linguistics and Graphs for Space

Artificial Intelligence Engineer & AI Product Owner

Dephion/Habtic/Yunify (Company name changed)

Feb 2019 - Sep 2022 | Geleen, the Netherlands

- Built and shipped a production recommender system within a consumer health app, directly powering the core personalization engine used by real users.
- Migrated the existing recommender framework into the new Databricks tech stack.
- Designed and ran an A/B testing framework for the app's users, generating empirical evidence for recommender model selection and product decisions.
- Developed NLP text mining tools to extract employee well-being signals from company reports, creating a novel internal analytics capability.
- Coordinated with other teams to get Data Science features into the company's products and roadmaps.

Artificial Intelligence and Data Science Consultant

iNutech (Part Time)

Jul 2020 - Jan 2022 | Brasilia, Brazil (Remote)

- Developed a NLP framework for extracting relevant legal information from legal proceedings from Brazilian courts, including named entity recognition and annotation of different judge decisions.
- Managed end-to-end data annotation pipeline with domain expert lawyers, ensuring high-quality labeled data for a sensitive, high-stakes legal domain.

Neural Information Retrieval Intern

Advanced Concepts Team - European Space Agency

Sep 2020 - Dec 2020 | Noordwijk, the Netherlands

- Selected for ESA's Advanced Concepts Team — the Agency's far-future research think tank — to explore document similarity using sentence and document embeddings.
- Applied NLP to measure idea novelty against ESA's historical study archive, and matched personnel competencies to submitted project proposals on ideas.esa.int.

Artificial Intelligence Intern

Accenture

Sep 2018 - Dec 2019 | Heerlen, the Netherlands

- Worked on thesis: Stock Portfolio Selection with Non-Stationary Contextual Multi-Armed Bandits.
- Worked with Machine Learning and Natural Language Processing for developing an IT ticket classification system.

Artificial Intelligence Intern

Center for Management and Strategic Studies (CGEE)

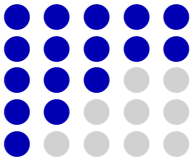
Aug 2014 - Aug 2016 | Brasilia, Brazil

- Increased data analysis capabilities by applying NLP techniques to build scientific publication networks
- Supported the evaluation of the National Institutes of Science and Technology public policy in Brazil

AWS | Databricks | SQL | Cypher | Docker | FastAPI | Embeddings

LANGUAGES

English
Portuguese
Spanish
Italian
Dutch



PUBLICATIONS

Conference Proceedings

- Berquand, A. [Audrey], Ladeira, A. V., & Dold, D. (2024, October). Lost in space but not in data: Tracking technology trends in the space field. doi:10.5281/zenodo.13885543
- Berquand, A. [A.], & Ladeira, A. V. (2022). From mission description to knowledge graph: Applying transformer-based models to map knowledge from publicly available satellite datasets. In *9th international systems & concurrent engineering for space applications conference digital event on 30 september - 2 october 2020*. European Space Agency.

Journal Articles

- Maia, J., Ladeira, A. V., Cagnin, C., & Villela, A. (2015). Análise de redes e fta para uma avaliação estratégica dos institutos nacionais de ciência e tecnologia. *Parcerias Estrategicas*, *40*(20).

OTHER PROJECTS

Honours+ - The Position of Women in I.T.

Maastricht University

Aug 2017 - Jun 2018 | Maastricht, NL

- Worked in a multidisciplinary group of other honours students to develop a set of recommendations for the board of Maastricht University to promote women in IT

UnBeatables Robotics Team

University of Brasilia

Sep 2015 - Jun 2016 | Brasilia, Brazil

- 1st place at RobotCup – Standart Platform League – Drop-In Only 2016
- 1st place at CBR/LARC (Brazilian Robotics Competition/Latin American Robotics Competition) – Standart Platform League – 2015